

KRAMFORS HOGA KUSTEN, Sweden

WMO: 022590

Lat: 63.049N

Lon: 17.769E

Elev: 34

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-9.0	-18.5	2.0	-13.5	-13.0	2.7	-8.5	24.3	31.8	21.5	30.6	4.3	0	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	78.7	64.2	75.2	62.6	71.7	60.8	66.6	75.6	64.3	72.8	62.2	69.3	9.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	85.8	69.4	60.9	79.9	67.6	59.1	74.9	65.6	31.1	75.6	29.5	73.3	28.0	69.7	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	17.8	15.7	-16.1	82.8	8.1	4.6	-21.9	86.1	-26.7	88.8	-31.2	91.4	-37.2	94.8		
			-17.4	67.4	7.7	3.2	-23.0	69.7	-27.5	71.6	-31.9	73.4	-37.5	75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	17.5	17.7	28.3	38.4	48.0	55.6	61.8	58.3	50.4	39.0	29.6	22.2
	DBStd	17.32	12.55	13.43	8.97	5.32	7.04	5.42	5.33	4.82	5.34	5.85	8.28	11.23
	HDD50	4949	1008	904	674	348	124	11	0	2	60	343	612	863
	HDD65	9527	1473	1324	1139	797	526	284	133	216	439	806	1062	1328
	CDD50	939	0	0	0	0	63	179	364	260	71	2	0	0
	CDD65	43	0	0	0	0	0	3	32	8	0	0	0	0
	CDH74	424	0	0	0	0	21	53	303	47	0	0	0	0
	CDH80	70	0	0	0	0	0	6	62	2	0	0	0	0
Wind	WSAvg	6.3	5.7	5.6	6.8	6.5	7.3	7.8	6.7	6.1	5.8	5.8	5.6	5.4
Precipitation	PrecAvg	24.4	1.8	1.1	1.2	1.4	1.6	2.1	3.5	3.1	2.3	2.2	2.1	1.9
	PrecMax	35.9	3.4	2.4	3.0	3.4	3.5	4.4	7.0	6.5	5.8	6.4	6.0	3.9
	PrecMin	17.0	0.2	0.2	0.2	0.2	0.0	0.9	0.4	0.9	0.7	0.5	0.8	0.3
	PrecStd	4.4	0.8	0.6	0.6	0.8	0.8	0.9	1.7	1.7	1.5	1.4	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	46.1	51.7	65.9	77.0	80.3	85.7	78.8	69.7	57.4	49.7	44.5
		MCWB	37.1	39.6	43.4	49.7	58.9	62.4	65.8	66.7	58.3	50.3	46.0	40.2
	2%	DB	37.6	42.8	46.7	59.3	71.5	74.8	80.9	75.0	65.8	53.7	44.5	40.9
		MCWB	35.7	38.1	39.3	47.4	56.9	60.6	65.7	63.2	57.1	49.1	43.2	37.8
	5%	DB	35.7	37.6	44.2	54.0	66.4	70.2	77.1	71.4	62.6	51.6	42.5	37.7
		MCWB	34.5	34.3	37.9	44.1	53.9	57.9	64.5	61.9	55.8	47.7	40.9	36.3
	10%	DB	33.8	34.2	40.6	50.2	62.3	66.5	73.5	68.1	60.4	48.4	40.6	35.8
		MCWB	33.1	33.1	35.4	41.4	51.6	56.0	62.7	59.6	54.8	45.9	39.3	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	40.2	43.8	50.8	60.9	64.7	69.4	68.6	61.2	52.7	46.6	40.2
		MCDB	39.2	44.8	50.3	64.0	71.6	75.2	76.9	74.9	65.8	56.0	48.5	42.5
	2%	WB	36.0	37.9	40.3	47.5	58.0	62.0	67.4	65.6	59.1	50.5	42.9	37.9
		MCDB	37.1	42.0	45.5	58.0	69.5	70.8	76.8	72.3	63.6	52.8	43.4	39.3
	5%	WB	34.2	35.1	37.8	44.7	55.2	60.0	65.5	63.0	57.3	48.1	40.6	36.2
		MCDB	35.1	36.5	42.6	53.3	64.8	67.7	75.1	69.0	61.3	50.4	41.2	37.2
	10%	WB	32.6	33.1	35.6	42.1	52.1	57.9	63.7	61.0	55.6	45.8	38.6	34.5
		MCDB	33.3	34.0	39.1	48.8	61.2	64.7	72.2	66.1	59.3	47.9	39.4	35.3
Mean Daily Temperature Range	5% DB	MDBR	12.7	16.1	17.3	18.3	19.9	17.7	18.4	16.9	16.1	12.9	9.7	11.9
		MCDBR	12.7	15.5	19.4	27.6	27.4	23.3	23.5	22.2	19.4	16.8	11.3	13.1
		MCWBR	11.5	12.4	13.2	16.1	14.6	12.0	11.2	12.5	12.5	12.7	9.8	11.7
	5% WB	MCDBR	12.2	13.6	17.3	26.3	25.2	20.4	21.3	18.4	16.8	14.9	10.6	12.2
		MCWBR	11.5	11.5	12.3	15.8	13.8	11.6	10.9	11.1	11.8	12.5	9.7	11.1
Clear-Sky Solar Irradiance	taub	0.233	0.293	0.317	0.357	0.357	0.353	0.369	0.360	0.328	0.312	0.375	0.348	
	taud	2.064	2.204	2.294	2.348	2.425	2.469	2.443	2.465	2.541	2.431	2.717	2.602	
	Ebn at Noon	127	203	248	262	272	276	267	259	248	200	119	72	
	Edh at Noon	10	20	27	31	31	31	31	28	21	16	9	6	
All-Sky Solar Radiation	RadAvg	56	252	717	1206	1542	1766	1613	1224	750	318	85	28	
	RadStd	7	43	80	103	165	116	168	104	85	55	17	3	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-221	N/A	N/A	N/A	

Nomenclature: See separate page